Dart: An Easy, Scalable and Multi-purpose Programming Language

If you are looking for an application programming language that is easy to learn, highly scalable and deployable everywhere, then do try Dart. This article introduces readers to its exciting array of features.

Dart is a programming language developed at Google. The primary developers behind it are Lark Bak and Kasper Lund. Dart was first released in October 2011, and the latest stable release is version 1.22, which came out in February 2017. The consistent updates to Dart indicate its active development.

Dart is cross-platform and supports all major operating systems. It is open source and available with a BSD licence. It is recognised as an ECMA standard.

Objectives of Dart

According to the official documentation, Dart is a long-term project with ambitious objectives. The core objectives, illustrated in Figure 1, are:

- Dart has a strong support base with many libraries and tools, which enable very large applications.
- One of the major objectives of Dart is to simplify programming tasks. It is designed to make common programming tasks simpler.
- Dart is very stable and it can be used to build production quality real-time applications. It is an object-oriented programming language with support for inheritance, interfaces and optional typing features.

Dart targets

The Dart code can be executed in four different ways (see Figure 2):

- The code can be trans-compiled into JavaScript using source-to-source compilation. This is done with the help of the dart2js compiler. As it can be converted to JavaScript, all major browsers support it.
- The Dart SDK includes a virtual machine called Dart VM. This mode is for standalone execution. The Dart code can be executed in command line interface mode. The Dart SDK includes a powerful package manager called 'pub'.
- The Dart code can be executed in another mode through the Dartium browser. This is a customised Chromium Web browser that includes the Dart VM. As this browser has direct support for Dart code, there is no need to convert it into JavaScript.
- Dart can also be executed in AOT mode. AOT stands for Ahead-Of-Time compilation. In this mode, the Dart code can be directly converted into native machine code.

DartPad

If you prefer to start writing your first Dart program without any installation or configuration, then DartPad is the right